

InceptionV3 in Medical Imaging: Enhancing Precision in Acute Lymphoblastic Leukaemia Diagnosis

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Abstract— Acute lymphoblastic leukaemia (ALL) is a group of blood malignancies that need very sophisticated detection methods and constitute a major danger to public health across the world. With the inceptionV3 model developed and implemented, this research study significantly advances blood cancer diagnostics. This model is built on top of a massive Blood Cells Cancer (ALL) dataset that includes a spectrum of cellular morphologies linked to ALL. This dataset and deep learning approaches were used to train the model being considered. A key component in enabling prompt and successful medical intervention, the major objective of this research activity is to enhance the precision and dependability of blood cancer diagnosis. Images of blood cells may reveal subtle aberrations and patterns, which may be recognised using a fine-tuned inceptionV3 model. The inceptionV3 model can use this to better differentiate between healthy and cancerous cells. The training procedure involves refining the model's parameters to reach better degrees of sensitivity and specificity. After the training phase is over, the suggested inception model achieves an astonishing accuracy rate of 98.46%, demonstrating an excellent level of performance. The model demonstrates remarkable accuracy in diagnosing blood cancer with a high degree of precision, highlighting its potential as a crucial diagnostic tool in clinical settings.

Keywords—InceptionV3, Acute Lymphoblastic Leukaemia, Deep Learning, Blood Cancer, CNN.

I. INTRODUCTION

Blood cancers, including acute lymphoblastic leukaemia (ALL) in particular, pose a serious challenge to international healthcare systems. A timely and accurate diagnosis is crucial for patients because of the complex biological symptoms and the potential for rapid progression associated with the ailment. This allows for the creation of effective treatment plans and improves patient outcomes. In this specific setting, state-of-the-art computational tools, such as InceptionV3, have opened up unique possibilities for enhancing the accuracy and efficacy of blood cancer diagnosis. These opportunities are now more accessible than ever before because to advancements in artificial intelligence. This work presents a thorough investigation into

the identification of haematological cancers, namely acute lymphoblastic leukaemia (ALL), by using the capabilities of a novel InceptionV3 architecture. One important reason why this effort is so important is the urgent need for reliable and fast diagnostic tools that may improve upon current approaches utilised in clinical settings. Traditional diagnostic methods often include the manual analysis of blood cell samples. There is a high potential for subjectivity and human mistake in this method. The aforementioned issues may be amenable to a data-centric approach that artificial intelligence and deep learning models may provide, which allows for the ultra-efficient and accurate analysis of massive datasets [1].

This study relies on the inceptionV3 model that has been fine-tuned using the Blood Cells Cancer dataset. This study was based on this model. The dataset includes a wide range of cellular morphologies, which allows it to provide a complete picture of the changes linked to Acute Lymphoblastic Leukaemia (ALL). In order to improve the accuracy of distinguishing between normal and malignant blood cells, the proposed model employs inceptionV3's natural ability to detect intricate patterns inside pictures. This ability enables inceptionV3 to evaluate and categorise visual content. The motivation for conducting the investigation that is the focus of this study comes from the desire to enhance current diagnostic methods and the hope of impacting patient outcomes via the promotion of quick and precise diagnosis. Achieving an exceptionally high level of accuracy in the detection of blood cancer is the main objective of the research effort, which involves fine-tuning the inceptionV3 model [2].

This work is intended to add to the existing body of knowledge on blood cancer diagnosis and the broader conversation around the use of AI in healthcare. Our investigation into the intricacies of the proposed inceptionV3 model and its application to the Acute Lymphoblastic Leukaemia dataset will enable us to achieve this goal. Acute Lymphoblastic Leukaemia (ALL) is the dataset that will be the primary emphasis of this investigation. This research aims to examine the limitations of conventional diagnostic

methods and propose alternatives that are more suited to the dynamic field of medical technology. Ideally, this will lead to diagnostic processes that are more precise and dependable [3].

II. LITERATURE REVIEW

The most current developments in the use of deep learning techniques for the diagnosis of acute lymphoblastic leukaemia (ALL) are reviewed in detail in this section. Study after study delves into the complex terrain of ALL detection, illuminating diverse approaches, algorithms, and models that shape medical image analysis's dynamic environment. The merits and shortcomings of each method are better understood by a comparison of various investigations, which provides useful information for both theoretical and practical considerations in the medical field.

With the use of artificial intelligence, R. Khandekar et al. [4] developed a novel method for detecting ALL by analysing pictures of blood smears taken at a microscopic level. Using the You Only Look Once (YOLOv4) technique, their system got great mean average accuracy scores of 96.06% on the ALL_IDB1 dataset and 90.77% on the C_NMC_2019 dataset. The capability of this AI technology to handle complicated picture data demonstrates its promise for pre-screening in ALL diagnosis. In their study, Eid et al. [5] presented a new method for identifying cancer cells in blood and skin using photonic crystal fibres (PCFs). With a frequency of 2.0 THz, the model showed sensitivity levels of over 96% for both forms of cancer, proving its effectiveness. A focus on practicality was highlighted by the sensor, which demonstrated potential for real-world applications with reduced material and confinement losses. S. Shafique et al. [6] established a system that can automatically identify ALL in microscopic blood pictures. Their method showed great sensitivity (92% confidence interval) and specificity (91% confidence interval) using the ALL-IDB-1 dataset, with a total accuracy of 93.7%. The diagnosis procedure is simplified with this approach, which provides a trustworthy automated way for ALL detection.

To automate the process of ALL detection using deep CNNs, C. Mondal et al. [7] put up a solution. Weighted F1-score: 88.6%, balanced accuracy: 86.2%, and AUC: 0.941 were the results obtained using the weighted ensemble model. An effective strategy for detecting ALL in pictures of tiny cells is shown here by combining deep learning with ensemble approaches. As an approach to ALL detection, S. Praveena and S. P. Singh [8] presented a Deep Convolutional Neural Network (CNN) trained using the Grey Wolf-based Jaya Optimisation Algorithm. With the use of colour histogram characteristics and sparse fuzzy C-Means clustering, their method demonstrated the ability to accurately analyse ALL and reached maximal accuracy, sensitivity, and specificity. An investigation of transfer learning models for acute lymphocytic leukaemia from blood smear cells was carried out by K. S. Ananthu et al. [9]. The findings showed that different models had notably high accuracy rates, which shed light on how different transfer learning methods compare.

The use of hypercomplex-valued convolutional neural networks (HvCNNs) for lymphocyte categorization in blood smear pictures was investigated by G. Vieira and M. Eduardo Valle [10], with a focus on its potential to assist in the detection of ALL. Picture processing in the HSV colour

space using Clifford algebras was where the HvCNN model really shone. This new way of thinking about ALL categorization has a 96.6% total success rate. Two researchers, S. Shafique and S. Tehsin, used deep convolutional neural networks to identify and categorise four distinct forms of ALL [11]. Their method, which made use of data augmentation methods and a pre-trained AlexNet model, achieved 96.06% overall accuracy, 99.03% specificity, and 96.74% sensitivity. In a surprising departure from the norm, the approach showed impressive accuracy even without microscopic picture segmentation. The summary of these reviews are shown in table I.

TABLE I. WORK DONE BY PREVIOUS AUTHORS

S. No.	Author Name	Model Name	Accuracy
1	R. Khandekar et al. [4]	YOLOv4	90.77%, 96.06%
2	S. Shafique et al. [6]	CNN	93.7%
4	C. Mondal et al. [7]	CNN	86.2%
5	S. Praveena and S. P. Singh [8]	CNN	93.50%
6	K. S. Ananthu et al. [9]	Transfer Learning	97.88%
7	G. Vieira and M. Eduardo Valle [10]	HvCNNs	96.6%
8	S. Shafique and S. Tehsin [11]	AlexNet	96.06%

Every one of the papers that were looked at adds something special to the field of ALL detection. An AI-powered pre-screening system was developed by R. Khandekar et al. [4] using the YOLOv4 algorithm, while another method with great sensitivity for skin and blood cancer cells was proposed by M. M. A. Eid et al. [5] using a PCF-based sensor. An automated approach by S. Shafique et al. [6] that streamlines the diagnosis procedure and achieves excellent accuracy with the ALL-IDB-1 dataset stands out. The ensemble model by C. Mondal et al. [7] demonstrates the power of weighted convolutional neural networks (CNNs), whereas the GreyJOA-trained CNN by S. Praveena and S. P. Singh [8] emphasises the significance of optimisation strategies. The comparative research by K. S. Ananthu et al. [9] sheds light on the advantages of various transfer learning models, whilst the application of HvCNNs by G. Vieira and M. Eduardo Valle [10] gives a fresh viewpoint with impressive accuracy rates. In conclusion, the method developed by Shafique and Tehsin [11] is notable for accurately classifying subtypes of images without using picture segmentation.

III. PROPOSED METHODOLOGY

An extensive and thorough description of the methods used throughout the investigation is provided in this section. We provide the dataset that our model was trained on and describe in detail the steps used to prepare it for analysis via

preprocessing. The proposed research method is shown in Fig. 1 as a simplified flow diagram. An overview of the research's sequential methods and processes is what the visual guide is trying to accomplish. It provides a thorough synopsis of the experimental design and the data processing procedure.

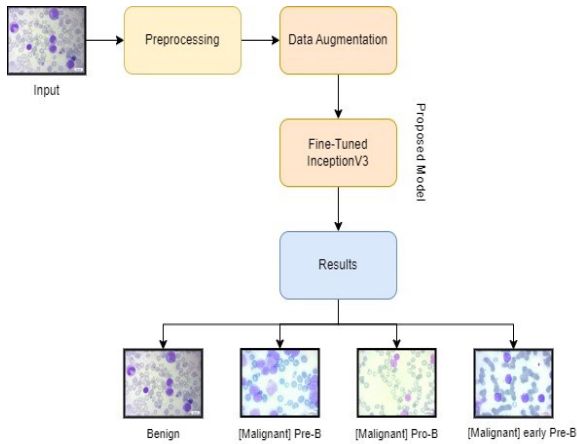


Fig.1 Proposed methodology

A. Dataset

Acute lymphoblastic leukaemia (ALL) is a very widespread cancer that necessitates accurate diagnosis due to the invasive, costly, and time-intensive nature of the condition. Research developing more effective screening procedures will potentially provide improved patient prognosis and outcomes. There are many non-invasive techniques for distinguishing between malignant and benign disorders, among which is the examination and interpretation of peripheral blood smear (PBS) pictures. This study utilises data obtained from the bone marrow unit at Taleqani Hospital in Tehran, Iran. In this particular dataset, the researchers selected a total of 3242 images of peripheral blood smears obtained from a sample of 89 individuals who were suspected to be afflicted with the condition under investigation. The source of this information is derived from the citation [12]. According to Figure 2, the dataset exhibits a categorization of malignant lymphoblasts into three distinct categories, namely Early Pre-B, Pre-B, and Pro-B Acute Lymphoblastic Leukaemia. The use of a Zeiss camera and a customised flow cytometry instrument enhances the reliability of the dataset. The absence of precise symptoms and indicators may lead to errors when laboratory professionals manually analyse PBS photographs. Given the aforementioned circumstances, it becomes evident that the use of advanced computational methodologies, such as deep learning models, is important in order to augment the diagnostic process. This approach would provide a systematic and impartial analysis of PBS images, hence mitigating the likelihood of erroneous diagnostic outcomes.

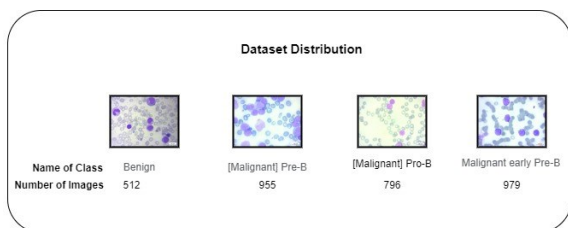


Fig.2. Dataset Distribution

B. Preprocessing and data augmentation

In the field of machine learning, the preprocessing step plays a crucial role since it entails transforming raw data into a format that is suitable for training models. The first stage of preparation provides the essential foundation for accurately and efficiently acquiring patterns within the dataset. In the field of computer vision tasks, such as image classification, data preparation involves a series of carefully defined activities aimed at ensuring optimal model performance [13].

Initially, the dataset undergoes an intentional procedure of partitioning, which is a fundamental practise within the domain of machine learning. The objective of this procedure is to distinguish the training set from the validation set. In the present scenario, the dataset is divided into a ratio of 9:1, where 90% of the samples are assigned to the training set and the remaining 10% are allocated to the validation set. The choice of this ratio is often determined with the intention of striking a balance between having a sufficient quantity of data for training and reserving a significant portion for evaluating the model's performance on unseen samples. The inclusion of the random state parameter, particularly set to a value of 88, ensures the reproducibility of outcomes. This feature enables the consistent generation of identical data partitions when the programme is ran many times. The paramount relevance is in guaranteeing consistency in results and facilitating smooth comparison across several trials. Following the process of division, the third step involves the modification of images to adhere to a standardised dimension. The selection of a target size of 224x224 pixels aligns with established norms in deep learning models that use image data [14]. Square input shapes are often used in these models because they provide simplicity and compatibility with frequently utilised architectures. The process of resizing ensures uniformity in the input data, hence allowing the model to effectively handle images with consistent dimensions throughout both training and evaluation phases. The technique outlined above not only enhances the efficiency of the computing operation but also enhances the model's ability to generalise across diverse datasets. Subsequently, the dataset undergoes a random shuffling procedure, so introducing an element of unpredictability in the arrangement of samples. The process of randomising the order of data instances aids in mitigating the risk of the model acquiring knowledge that is specific to the sequential organisation of the data. Without the process of shuffling, there is a potential for the model to accidentally develop correlations that are connected to the specific order in which the samples were presented during the training phase. This has the potential to compromise its ability to properly transfer acquired knowledge to novel and unfamiliar information. The usage of the random shuffling approach serves to disturb any inherent structure present in the dataset, hence promoting a learning process that is characterised by increased resilience and impartiality [15].

C. Model

The use of the InceptionV3 model in this research study signifies a deliberate and calculated choice, driven by the model's robust architectural features and established track record of success in diverse picture categorization assignments. InceptionV3, an architectural design for deep

convolutional neural networks (CNNs), was created by Google. This network has shown remarkable efficacy in the tasks of feature extraction and classification, especially when applied to extensive visual recognition difficulties [16]. The InceptionV3 model is well recognised for its distinctive architectural design, which is distinguished by the integration of inception modules. The use of several convolutional processes with varying kernel sizes inside a single layer enables the model to effectively collect information at different scales concurrently. The InceptionV3 model has the capacity to process information at several scales, which allows it to successfully analyse complicated patterns and fine details seen in medical pictures. This characteristic makes it particularly suitable for the intricate nature of detecting Acute Lymphoblastic Leukaemia (ALL). The architectural design also incorporates the use of global average pooling, which effectively decreases the spatial dimensions of the feature maps. Consequently, this reduction in spatial dimensions leads to a decrease in the number of parameters present in the fully linked layers. This approach not only reduces the occurrence of overfitting but also improves the model's capacity to make accurate predictions on new, unknown data. The InceptionV3 model adds batch normalisation, a technique that normalises the activations of the network. This inclusion enhances training stability and expedites convergence during the optimisation phase. The InceptionV3 model has exceptional performance in hierarchical feature extraction, enabling it to autonomously acquire and comprehend complex patterns seen in medical pictures. The variability in the morphology of blood cells is of utmost importance in the context of all detection. The use of inception modules facilitates the model's ability to dynamically extract features at various sizes, hence yielding a complete representation of the input data. The use of inception modules inside the InceptionV3 architecture serves to augment parameter efficiency. By integrating several kernel sizes inside a single layer, the model attains a harmonious equilibrium between the size of the receptive field and computing efficiency. The aforementioned characteristic is especially beneficial in the field of medical image analysis, since datasets may have constraints in terms of size, and the effective utilisation of parameters is crucial for optimising model performance. The InceptionV3 model is often used as a pre-trained model on large picture datasets, such as ImageNet [17]. By using transfer learning, the model is able to acquire information obtained from many image recognition tasks, enabling it to effectively apply this knowledge to the particular characteristics that are pertinent to ALL detection, resulting in improved generalisation capabilities. This particular feature has significant value in situations when there is a scarcity of medical imaging datasets. The use of global average pooling and batch normalisation techniques in InceptionV3 enhances the model's resilience and ability to generalise. By mitigating the issue of overfitting and enhancing the stability of the training process, the model becomes more proficient in accommodating fluctuations in the attributes of blood cell pictures, hence assuring consistent and dependable performance over a wide range of scenarios. The InceptionV3 model has strong suitability for multi-class classification tasks, making it a suitable option for the

categorization of several subtypes of Acute Lymphoblastic Leukaemia (ALL) or the differentiation between malignant and benign cells. The exact classification of varied classes is of utmost importance in medical diagnostics, making it vital for the model to exhibit great accuracy in handling such cases. The use of InceptionV3 within the domain of acute lymphoblastic leukaemia (ALL) detection capitalises on its inherent capabilities to address the complexities and diversities inherent in blood cell imagery. The model's capacity to extract hierarchical features, together with its efficient use of parameters and its ability to transfer learning, corresponds to the requirements of medical picture analysis. The use of global average pooling and batch normalisation techniques improves the stability and flexibility of the model, enabling it to effectively handle diverse datasets. Consequently, the model consistently delivers dependable performance in accurately identifying instances of acute lymphoblastic leukaemia (ALL)[18].

IV. RESULTS

The effectiveness and efficiency of a deep learning model are the primary objectives of this research. The study delves deep into the methodology used to train and assess the model.

Thirty epochs, or complete iterations of the whole dataset, made up the training methodology that the InceptionV3 model underwent. A deliberate choice of epochs is made to achieve a fine balance between avoiding the risk of overfitting and letting the model learn the intricate patterns in the training data. The learning rate, set at a value of 0.0001, plays a crucial role in directing the optimisation process throughout the training phase. To reduce the likelihood of overshooting and oscillation in the optimisation landscape, a modest learning rate was used so that the model could converge gradually. Medical image analysis is a domain where extreme caution is required due to the complexity of the datasets used, which necessitates a more nuanced and careful convergence. The achieved accuracies are a reflection of the results of the meticulous training programme. These accuracies demonstrate how well the model can classify the training and validation datasets. The astounding training accuracy of 99.96% (Fig.3) demonstrates the model's ability to learn from the training data and understand the complex relationships within the images of blood cells. Based on the observed high training accuracy, it seems that the model has successfully adapted to the complicated and detailed training dataset, proving that it can differentiate between benign and malignant cells. The validation accuracy, which peaked at 99.66% after 30 epochs, is equally important. The model's ability to generalise its performance to new data is strongly shown by the aforementioned statistic. To test how well the model can generalise its training data to new situations, it is used the validation dataset. A validation accuracy of 99.66% (Fig.4) shows that the model is good at learning from the training data and also performs well when given new images of blood cells. The medical image analysis sector stands to benefit greatly from the established validation accuracy level, since accurate categorization of unseen occurrences is essential for real-world application. Another encouraging sign is the apparent parity between the training and validation sets' accuracy levels. If these two

metrics diverge, it might be a sign of overfitting, a problem where the model becomes overly specific to the features of the training data and struggles to apply its knowledge to new situations. An impressive alignment between the model's training accuracy of 99.96% and validation accuracy of 99.66% suggests that the model has accomplished its dual goal of learning from the training set and generalising effectively to new instances. The occurrence of this harmony bodes well for the model's applicability in the actual world, where it is inevitable to encounter a wide range of hitherto unseen blood cell images [19]. The inceptionV3 model was rigorously tested using a dedicated test dataset after the training process. As demonstrated in table II, it proved to be resilient and successful in classifying Acute Lymphoblastic Leukaemia, with an exceptional accuracy of 99.38%.

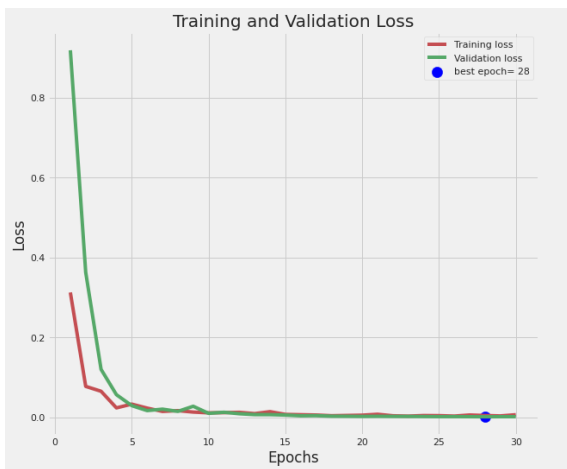


Fig.3. loss graphs.

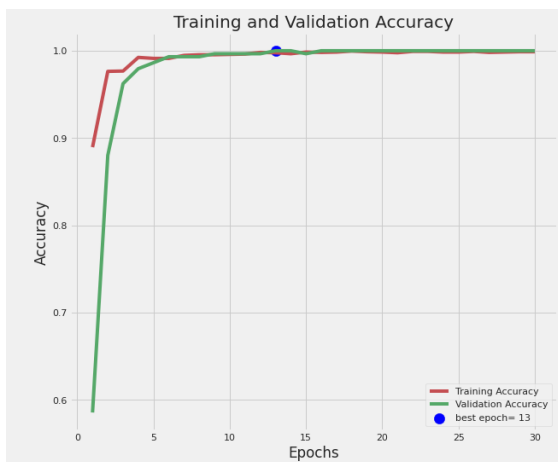


Fig.4. Accuracy graphs.

Table II. Performance Matrix

Classes	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Benign	100	98	99	99.38
Malignant Pre-B	100	100	100	
Malignant Pro-B	100	100	100	
Malignant early Pre-B	96	100	98	

A confusion matrix is a crucial tool in machine learning and classification tasks, providing a detailed breakdown of a model's predictions and their comparison to actual outcomes. It helps evaluate model performance, class imbalance, error analysis, threshold adjustment, model selection and comparison, medical or anomaly detection, business strategy adjustment, and hyperparameter optimization. It aids stakeholders in making informed decisions, optimizing models, and assessing their suitability for real-world applications [20]. The confusion matrix is shown in Fig.5.

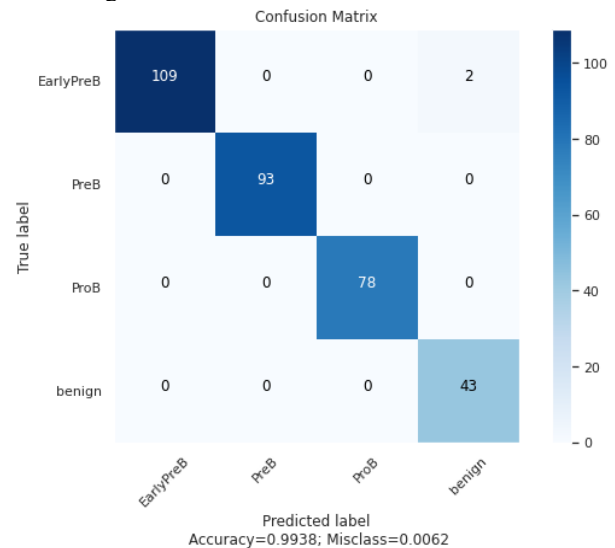


Fig.5. Confusion Matrix

V. CONCLUSION

This work presents a fine-tuned inceptionV3 model that has been particularly created for the purpose of detecting blood cancer. The efficacy of this paradigm was shown to be very high. Following a rigorous training process consisting of 30 epochs and with a learning rate of 0.0001, the model exhibited exceptional performance. The model attained a training accuracy of 99.96% and a validation accuracy of 99.66%. Both of these percentages exhibit a significant deviation from what would be expected by chance alone. The model's ability to capture intricate patterns while yet maintaining its ability to generalise to novel and uncharted scenarios is seen in its potential to create a harmonious alignment between the accuracy of its training and validation. The model demonstrated a high level of accuracy, namely 99.38%, throughout extensive testing on a specialised dataset. This result provides evidence of the model's robustness when used in practical situations. The model's ability to differentiate between benign and malignant blood cells may be partially ascribed to the incorporation of convolutional and separable convolutional layers, along with the use of batch normalisation, dropout mechanisms, and meticulous hyperparameter tweaking. This work emphasises the potential of sophisticated inceptionV3 architectures in enhancing diagnostic capacities in the field of blood cancer detection. It contributes to the growing corpus of research that use deep learning for the interpretation of medical images. The positive results shown in this study provide promising prospects for the future development and use of these models within clinical environments. This resource

will serve as a valuable tool for healthcare practitioners, facilitating the correct and prompt identification of cancer.

Future research in this field should investigate the possible synergies among various techniques, with the aim of integrating the advantageous aspects of multiple models to develop more resilient and precise systems for the identification of Acute Lymphoblastic Leukaemia (ALL). Furthermore, it is important to focus on tackling obstacles such as the unpredictability of datasets and the interpretability of models. This will guarantee the smooth integration of these sophisticated approaches into clinical workflows, ultimately leading to improved diagnosis accuracy in the field of Acute Lymphoblastic Leukaemia.

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